

Vectors and Vector Spaces

Part 2/3: Linear Combinations and Spanning Sets

CSC 1105 Mathematics for Computer Science

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Lecture Outline

- Review of vector operations and subspaces
- Linear combinations and their coefficients
- Spans, spanning sets and geometric interpretation
- Matrix methods for testing membership and spanning
- Introduction to bases, coordinates and dimension
- Applications and exercises

Learning Outcomes

By the end of this lecture, you should be able to:

- Express a vector as a linear combination of given vectors.
- Describe the span of a finite set of vectors.
- Use a linear system to test whether a vector belongs to a span.
- Determine whether a set spans $\mathbb{R}^n$ or a specified subspace.
- Identify simple bases and use them to describe coordinates and dimension.

Review of Part 1

For vectors in the same coordinate space, addition and scaling act componentwise:

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

A subspace W of a real vector space V :

- contains the zero vector,
- is closed under vector addition,
- is closed under multiplication by every real scalar.

Applications in Computer Science

- **Computer graphics:** Basis vectors define coordinate systems. A position is represented as a linear combination of these vectors.
- **Signal and image processing:** Signals and images are represented using combinations of basis patterns. Selected patterns can provide compact approximations for compression.
- **Machine learning:** The span of feature vectors describes the representations they can produce. Linear dependence reveals redundant features.
- **Linear models:** Predictions take the form $\hat{\mathbf{y}} = X\mathbf{w}$, a linear combination of the columns of X . Exact fitting is possible only when the target lies in their span.

Constructing Vectors

Suppose the available vectors are

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Can we construct $\mathbf{w} = (3, 2)^T$ by scaling and adding them?

$$a\mathbf{u} + b\mathbf{v} = \begin{bmatrix} a + b \\ b \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Thus $b = 2$ and $a = 1$, giving

$$\boxed{\mathbf{w} = \mathbf{u} + 2\mathbf{v}.}$$

Which other vectors can these two vectors construct?

Linear Combinations

Definition

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ belong to a real vector space V . A linear combination of these vectors is any vector of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k, \quad a_1, \dots, a_k \in \mathbb{R}.$$

The scalars $a_1, \dots, a_k$ are the **coefficients**.

Coefficients may be positive, negative or zero. Ordinary linear combinations impose no requirement that the coefficients sum to 1.

Evaluating a Linear Combination

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

First substitute and scale:

$$2\mathbf{u} - \mathbf{v} + 3\mathbf{w} = \begin{bmatrix} 2(1) \\ 2(2) \\ 2(-1) \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 3(2) \\ 3(-1) \\ 3(1) \end{bmatrix}.$$

Then combine corresponding components:

$$= \begin{bmatrix} 2 - 0 + 6 \\ 4 - 1 - 3 \\ -2 - 3 + 3 \end{bmatrix} = \boxed{\begin{bmatrix} 8 \\ 0 \\ -2 \end{bmatrix}}.$$

Finding the Coefficients

Express $\mathbf{w} = (5, 4)^T$ as $a\mathbf{u} + b\mathbf{v}$, where

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Equating components gives

$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \iff \begin{cases} a + 2b = 5, \\ 2a + b = 4. \end{cases}$$

Subtracting twice the first equation from the second gives $-3b = -6$.

$$b = 2, \quad a = 5 - 2(2) = 1.$$

Therefore $\boxed{\mathbf{w} = \mathbf{u} + 2\mathbf{v}}$.

When a Linear Combination Is Impossible

Can $\mathbf{w} = (3, 5)^T$ be constructed from

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}?$$

The required equations are

$$a + 2b = 3, \quad 2a + 4b = 5.$$

Twice the first equation gives $2a + 4b = 6$, contradicting the second.

Conclusion

No real coefficients a, b produce $\mathbf{w}$. Every combination of $\mathbf{u}$ and $\mathbf{v}$ has its second component equal to twice its first.

When the Coefficients Are Not Unique

Using the same vectors $\mathbf{u} = (1, 2)^T$ and $\mathbf{v} = (2, 4)^T$, consider $\mathbf{w} = (3, 6)^T$.

$$\mathbf{w} = 3\mathbf{u} + 0\mathbf{v} = \mathbf{u} + \mathbf{v} = -\mathbf{u} + 2\mathbf{v}.$$

All solutions satisfy $a + 2b = 3$. With $b = t$,

$$a = 3 - 2t, \quad t \in \mathbb{R}.$$

Note

Existence asks whether any coefficients work. Uniqueness asks whether exactly one choice of coefficients works.

Linear Combinations: Exercise

Let $\mathbf{u} = (1, 1)^T$ and $\mathbf{v} = (1, -1)^T$.

- 1 Compute $2\mathbf{u} - 3\mathbf{v}$.
- 2 Find a, b such that $a\mathbf{u} + b\mathbf{v} = (4, 2)^T$.
- 3 For an arbitrary vector $(x, y)^T$, solve

$$a\mathbf{u} + b\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

- 4 Can these two vectors construct every vector in $\mathbb{R}^2$?

The Span of a Set of Vectors

Definition

The span of $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all linear combinations of its vectors:

$$\text{span}(S) = \left\{ \sum_{i=1}^k a_i \mathbf{v}_i : a_i \in \mathbb{R} \right\}.$$

- A linear combination with specified coefficients gives one vector.
- The span contains every vector obtainable as the coefficients vary.
- $\mathbf{w} \in \text{span}(S)$ means that at least one choice of coefficients produces $\mathbf{w}$.

Span of One Vector

For $\mathbf{u} = (1, 2)^T$,

$$\text{span}\{\mathbf{u}\} = \left\{ t \begin{bmatrix} 1 \\ 2 \end{bmatrix} : t \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} t \\ 2t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

This is the entire line $y = 2x$ through the origin.

For example, it contains $(0, 0)^T$, $(1, 2)^T$, $(-1, -2)^T$ and $(1/2, 1)^T$.

Special case

$\text{span}\{\mathbf{0}\} = \{\mathbf{0}\}$. A nonzero vector spans a line, whereas the zero vector spans only the zero subspace.

Span of Two Vectors in Three Dimensions

Let $\mathbf{u} = (1, 0, 1)^T$ and $\mathbf{v} = (0, 1, 1)^T$.

$$a\mathbf{u} + b\mathbf{v} = \begin{bmatrix} a \\ b \\ a + b \end{bmatrix}.$$

Therefore every vector in their span satisfies $z = x + y$.

Conversely, if $z = x + y$, then

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Hence

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \{(x, y, z)^T : z = x + y\}.$$

This is a plane through the origin in $\mathbb{R}^3$.

Geometric Possibilities for Spans

Vectors in $\mathbb{R}^3$	Their span
Only zero vectors	The zero subspace
One nonzero vector	A line through the origin
Two nonzero scalar multiples	The same line
Two vectors that are not scalar multiples	A plane through the origin
Three vectors with one outside the plane spanned by the other two	All of $\mathbb{R}^3$

Counting listed vectors alone does not determine the size of their span.

Why Every Span Is a Subspace

Let $W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$.

① Taking every coefficient to be zero gives $\mathbf{0} \in W$.

② If $\mathbf{u} = \sum a_i \mathbf{v}_i$ and $\mathbf{w} = \sum b_i \mathbf{v}_i$, then

$$\mathbf{u} + \mathbf{w} = \sum_{i=1}^k (a_i + b_i) \mathbf{v}_i \in W.$$

③ For any scalar c ,

$$c\mathbf{u} = \sum_{i=1}^k (ca_i) \mathbf{v}_i \in W.$$

Thus W passes the subspace test from Part 1.

Span as the Smallest Containing Subspace

Suppose a subspace U contains every vector in S .

Because U is closed under scaling and addition, it must also contain every linear combination of vectors in S .

$$S \subseteq U \implies \text{span}(S) \subseteq U.$$

Meaning of smallest

$\text{span}(S)$ is contained in every subspace that contains S . It is the smallest such subspace with respect to set inclusion.

For example, the smallest subspace containing $(1, 2)^T$ is the line $y = 2x$.

Spanning Sets

Definition

A set $S \subseteq V$ spans V if every vector in V is a linear combination of vectors in S . Equivalently, $\text{span}(S) = V$.

The standard coordinate vectors in $\mathbb{R}^3$ are

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$ for every $x, y, z \in \mathbb{R}$, these vectors span $\mathbb{R}^3$.

Matrix Form of a Linear Combination

Place the given vectors in the **columns** of a matrix:

$$A = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_k \\ | & | & \cdots & | \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}.$$

Then

$$A\mathbf{c} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k.$$

Membership test

$\mathbf{w} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ exactly when $A\mathbf{c} = \mathbf{w}$ has a solution.

If the vectors lie in $\mathbb{R}^n$, then A is $n \times k$ and $\mathbf{c} \in \mathbb{R}^k$.

A Span Membership Test: Example

Let $\mathbf{u} = (1, 0, 1)^T$, $\mathbf{v} = (0, 1, 1)^T$ and $\mathbf{w} = (2, -1, 1)^T$.

$$a\mathbf{u} + b\mathbf{v} = \mathbf{w} \iff \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Row-reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array} \right].$$

Thus $a = 2$, $b = -1$ and $\boxed{\mathbf{w} = 2\mathbf{u} - \mathbf{v}}$.

A Vector Outside the Span: Example

Keep $\mathbf{u} = (1, 0, 1)^T$ and $\mathbf{v} = (0, 1, 1)^T$, but take $\mathbf{q} = (2, -1, 4)^T$.

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{array} \right].$$

The last row requires $0a + 0b = 3$, which is impossible.

$$\boxed{\mathbf{q} \notin \text{span}\{\mathbf{u}, \mathbf{v}\}.$$

The geometric check agrees: $4 \neq 2 + (-1)$, so $\mathbf{q}$ is outside the plane $z = x + y$.

Testing Whether Columns Span All of the Space

For $A \in \mathbb{R}^{n \times k}$, its columns span $\mathbb{R}^n$ exactly when

$$A\mathbf{c} = \mathbf{w} \quad \text{is solvable for every } \mathbf{w} \in \mathbb{R}^n.$$

Row-reduction criterion

The columns span $\mathbb{R}^n$ if and only if a row-echelon form of A has a pivot in every row. Equivalently, $\text{rank}(A) = n$.

A pivot is the first nonzero entry in a nonzero row of row-echelon form. Rank is the number of pivots.

For a square matrix, this is also equivalent to $\det(A) \neq 0$.

Spanning the Whole Plane: Example

Do the vectors $\mathbf{u} = (1, 2)^T$ and $\mathbf{v} = (2, 1)^T$ span $\mathbb{R}^2$?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}.$$

There is a pivot in each of the two rows, so the columns span $\mathbb{R}^2$.

We can also construct arbitrary $(x, y)^T$ explicitly:

$$a + 2b = x, \quad 2a + b = y \quad \implies \quad a = \frac{2y - x}{3}, \quad b = \frac{2x - y}{3}.$$

These coefficients work for every real x, y .

Finding a Spanning Set from an Equation

Consider the subspace from Part 1:

$$W = \{(x, y, z)^T : x + y + z = 0\}.$$

Solve for x : $x = -y - z$. Let $y = s$ and $z = t$. Then

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Every vector in W has this form, and each displayed generator lies in W .

$$W = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

A Spanning Set for a Kernel

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix}$. To describe $\ker(A)$, solve

$$x + 2y - z = 0, \quad 2x + 4y - 2z = 0.$$

The second equation is twice the first. With $y = s$ and $z = t$,

$$x = -2s + t, \quad \mathbf{x} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Thus

$$\ker(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Free variables become coefficients of the spanning vectors.

Redundant Vectors in a Spanning Set

Suppose $\mathbf{w} = \mathbf{u} + \mathbf{v}$. Then

$$a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = (a + c)\mathbf{u} + (b + c)\mathbf{v}.$$

Adding $\mathbf{w}$ therefore produces no vectors beyond those already available:

$$\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u}, \mathbf{v}\}.$$

For example,

$$\text{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\} = \mathbb{R}^2.$$

The third vector is redundant because it is a combination of the first two.

Removing Redundancy Using Pivot Columns

Let $\mathbf{u} = (1, 0, 1)^T$, $\mathbf{v} = (0, 1, 1)^T$, $\mathbf{w} = (1, 1, 2)^T$.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The column space $\text{Col}(A)$ is the span of the columns of A . The pivot columns are 1 and 2. Here $\mathbf{w} = \mathbf{u} + \mathbf{v}$, so

$$\text{Col}(A) = \text{span}\{\mathbf{u}, \mathbf{v}\}.$$

Which columns to select

Use the pivot positions to select columns from the original matrix. Row operations generally change the column space itself.

Introduction to Linear Independence

A list $\mathbf{v}_1, \dots, \mathbf{v}_k$ is **linearly independent** if

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0} \implies a_1 = \dots = a_k = 0.$$

If a nonzero choice of coefficients gives zero, the list is **linearly dependent**.

For example, with $\mathbf{w} = \mathbf{u} + \mathbf{v}$,

$$\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}$$

uses coefficients $(1, 1, -1)$, so these three vectors are dependent.

Part 3 develops independence tests and their consequences in more detail.

Definition

A basis of a vector space V is a linearly independent spanning set for V .

- **Spanning:** the vectors can construct every vector in V .
- **Independence:** the representation has no redundant generators.

For example, $\{(1, 0)^T, (0, 1)^T\}$ is a basis of $\mathbb{R}^2$.

Adding $(1, 1)^T$ preserves spanning but destroys independence, so the enlarged set is not a basis.

A Nonstandard Basis: Example

Consider $\mathbf{u} = (1, 1)^T$ and $\mathbf{v} = (1, -1)^T$.

- They span $\mathbb{R}^2$, since we already showed

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{x+y}{2}\mathbf{u} + \frac{x-y}{2}\mathbf{v}.$$

- If $a\mathbf{u} + b\mathbf{v} = \mathbf{0}$, then

$$a + b = 0, \quad a - b = 0 \quad \implies \quad a = b = 0.$$

Thus $\{\mathbf{u}, \mathbf{v}\}$ is a basis of $\mathbb{R}^2$.

A vector space can have many different bases.

Coordinates Relative to an Ordered Basis

Let $B = (\mathbf{u}, \mathbf{v})$, with $\mathbf{u} = (1, 1)^T$ and $\mathbf{v} = (1, -1)^T$.

For $\mathbf{w} = (4, 2)^T$, we found $\mathbf{w} = 3\mathbf{u} + \mathbf{v}$. Its coordinate vector relative to B is

$$[\mathbf{w}]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Reconstruct the original vector using the basis columns:

$$\mathbf{w} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} [\mathbf{w}]_B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

The coordinate values depend on the chosen basis and its order.

Why Basis Coordinates Are Unique

Suppose a vector has two representations in a basis $B = (\mathbf{v}_1, \dots, \mathbf{v}_k)$:

$$\mathbf{w} = \sum_{i=1}^k a_i \mathbf{v}_i = \sum_{i=1}^k b_i \mathbf{v}_i.$$

Subtract the two expressions:

$$\sum_{i=1}^k (a_i - b_i) \mathbf{v}_i = \mathbf{0}.$$

Independence forces every coefficient to vanish:

$$a_i - b_i = 0 \quad \text{for all } i.$$

Therefore $a_i = b_i$ for all i : the basis coordinates are unique.

Dimension

Definition

For a finite-dimensional vector space V , its dimension is the number of vectors in a basis of V . Every basis of V has the same number of vectors.

Space	Dimension
$\mathbb{R}^n$	n
A line through the origin	1
A plane through the origin	2
The zero subspace $\{\mathbf{0}\}$	0

The zero subspace has the empty basis. The set $\{\mathbf{0}\}$ is not a basis because it is dependent.

Subspace Dimension and Ambient Dimension

Recall

$$W = \{(x, y, z)^T : x + y + z = 0\} = \text{span}\{(-1, 1, 0)^T, (-1, 0, 1)^T\}.$$

The two generators are independent: if

$$a \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \mathbf{0},$$

then the second and third components give $a = 0$ and $b = 0$.

$$\dim(W) = 2, \quad W \subseteq \mathbb{R}^3.$$

Each vector has three components, but only two coordinates relative to this basis are needed to describe it.

Polynomials as Linear Combinations

In $P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$, every polynomial is a combination of $1, x, x^2$.

$$p(x) = 2 - 3x + 4x^2 = 2(1) - 3(x) + 4(x^2).$$

The identity $a + bx + cx^2 \equiv 0$ requires $a = b = c = 0$, so these polynomials are independent.

$$B = (1, x, x^2), \quad [p]_B = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}, \quad \dim(P_2) = 3.$$

Connection to interpolation

Finding an interpolating polynomial amounts to finding coefficients in a polynomial representation subject to specified data values.

Application: Reconstructing a Short Signal

Use two signal patterns:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}.$$

Their combinations have the form

$$a\mathbf{u} + b\mathbf{v} = \begin{bmatrix} a + b \\ a - b \\ a + b \\ a - b \end{bmatrix}.$$

For $\mathbf{s} = (5, 1, 5, 1)^T$, solve $a + b = 5$, $a - b = 1$ to obtain $a = 3$, $b = 2$.

With these patterns fixed, two coefficients reconstruct this four-sample signal exactly. Signals outside their span need additional patterns or an approximation.

Revision Questions

- ① For $\mathbf{u} = (1, 0, 1)^T$ and $\mathbf{v} = (0, 1, 1)^T$, determine whether each vector belongs to their span:

$$\mathbf{p} = (3, 2, 5)^T, \quad \mathbf{q} = (3, 2, 6)^T.$$

- ② Find a spanning set for $W = \{(x, y, z)^T : x - 2y + z = 0\}$ and determine its dimension.
- ③ Does $\{(1, 0)^T, (0, 1)^T, (2, 3)^T\}$ form a basis of $\mathbb{R}^2$? Justify your answer.
- ④ Find the coordinates of $(6, 2)^T$ relative to $B = ((1, 1)^T, (1, -1)^T)$.

Part 2 Review and Part 3 Preview

- Linear combinations construct vectors by scaling and adding.
- The span is the subspace containing all such combinations.
- Solving $A\mathbf{c} = \mathbf{w}$ tests span membership.
- A basis gives unique coordinates, and its size gives dimension.

Part 3

We will develop linear dependence and independence further, then study dot products, orthogonality and related computations.

Questions?